



**ALLIANCE
UNIVERSITY**
Private University established in Karnataka State by Act No. 34 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi.

**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.:

**Title: Sales Performance Analysis Across Products, Categories &
Regions**

Submitted By

Name: Adarsh Kasaudhan

Reg. No.:2411021240076

Date:31-08-2026

[GITHUB LINK](#)

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 3

Create an interactive dashboard to analyze sales performance and identify key business patterns.

Problem Statement

A retail company maintains its sales data in the Sample Superstore dataset containing orders, products, customers, categories, regions, and profit information. The objective is to import the dataset into Power BI, clean and transform the data, create DAX measures, and develop an interactive dashboard that helps management understand sales performance across products, categories, and regions.

Step 1: Data Collection

- Imported the Sample Superstore CSV dataset.
- The dataset contains sales, profit, quantity, customer, product, category, and regional information.

Step 2: Data Import

- Opened Power BI Desktop.
- Selected Get Data → Text/CSV.
- Loaded the Superstore dataset into Power BI.

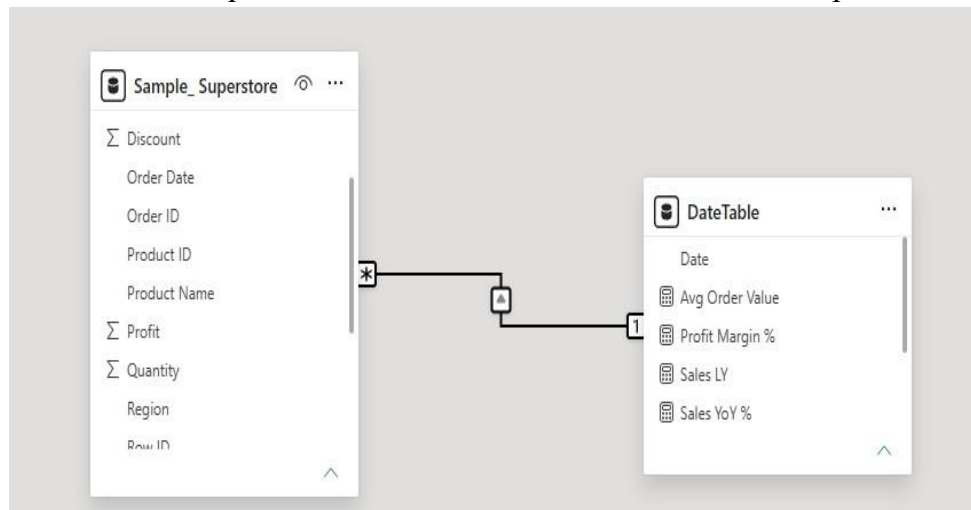
Step 3: Data Cleaning & Transformation

- Verified data types (Date, Text, Decimal Number).
- Removed duplicate and null records.
- Renamed columns where necessary.
- Converted Order Date into Date format.
- Created Month and Year fields using Power Query.

Step 4: Data Modelling

A single-table model was used because the Superstore dataset already contains integrated business information.

These relationships enabled accurate calculation of revenue and product-wise analysis.



Step 5: Dashboard Development

An interactive dashboard was created using Power BI with the following visuals:

- Designed an interactive Sales Dashboard in Power BI.
- Created KPI cards for Total Revenue, Average Price, Average Order Value, and Total Sold.
- Developed visualizations including bar, line, donut, and horizontal bar charts.
- Applied a professional theme with consistent colors, formatting, and layout.
- Enabled cross-filtering between visuals for dynamic data exploration.
- Validated DAX measures to ensure accurate business reporting.
- Published the final dashboard for business performance analysis and decision-making

DAX

Total Sales = SUM(Superstore[Sales])

Total Profit = SUM(Superstore[Profit])

Total Orders = DISTINCTCOUNT(Superstore[Order ID])

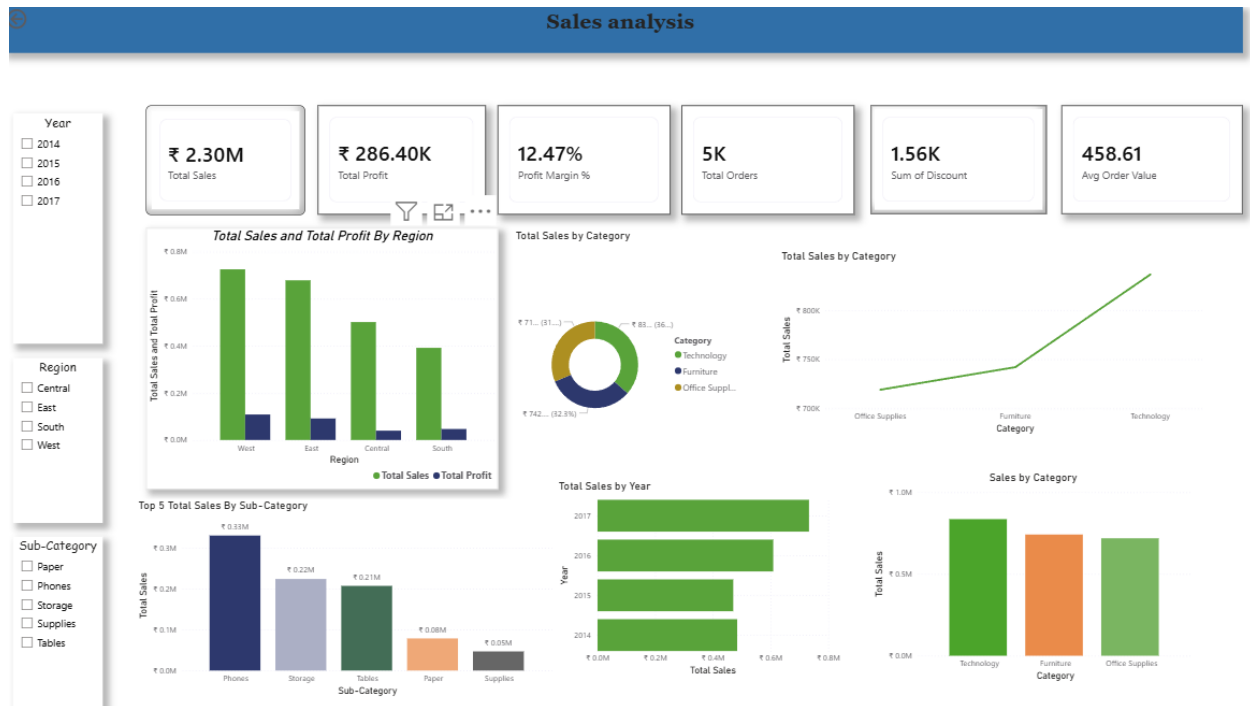
Total Quantity = SUM(Superstore[Quantity])

Average Sales = AVERAGE(Superstore[Sales])

Profit Margin % =

DIVIDE([Total Profit], [Total Sales], 0) * 100

Output



The developed Sales Dashboard provides:

- Total Revenue Performance
- Monthly Revenue Trend
- Category Analysis
- Size Revenue Distribution
- Top 10 Best-Selling
- Interactive filtering using subcategory, region, year

Business Insights

- Technology generated the highest overall sales revenue.
- The West region contributed the largest share of sales.
- Phones and Chairs were the best-selling product categories.
- Monthly sales showed noticeable fluctuations, indicating seasonal demand.
- Interactive slicers enabled dynamic analysis by region, category, and date.

Conclusion

This experiment successfully transformed the Sample Superstore dataset into an interactive Power BI dashboard. Data cleaning, transformation, DAX calculations, and visualization techniques were applied to generate meaningful business insights. The dashboard provides management with a centralized view of sales performance and supports effective data-driven decision making..

Learning Outcomes

Through this experiment, I learned:

- Importing CSV/Excel data into Power BI
- Data cleaning using Power Query
- Creating DAX measures for business KPIs
- Designing interactive dashboards
- Analysing sales across products, categories, and regions
- Generating actionable business insights from retail data